PROJECT PLANNING & MANAGEMENT

Urban Intelligence and Decision Support Dashboard for Egypt's New Administrative Capital

1. Project Proposal

Overview

This project proposes an interactive Microsoft Power BI dashboard that organizes data about Egypt's New Administrative Capital to support analysis of development patterns, geographical differences, potential service gaps, and possible interventions.

The dashboard will contain five analytical areas:

(1) City Overview

(2) Transportation and Mobility

(3) Environment and Sustainability

(4) Public Services

(5) Economy and Administrative Activity.

The project process includes data source research, acquisition, cleaning, transformation, modeling, KPI development, exploratory and geographical analysis where feasible, visualization, testing, and documentation.

Main Objective

The primary objective is to develop a functional, evidence based urban analytics dashboard that supports informed decision making by enabling users to explore urban indicators, compare geographical areas, identify potential service gaps, and examine data driven insights. The dashboard will integrate verified real world data with AI generated or AI assisted content where appropriate, while clearly distinguishing between observed data, estimated values, and simulated scenarios.

Specific Objectives and Success Criteria

- Data transparency: Document 100% of data sources, indicator definitions, calculation methods, and relevant assumptions.
- Data quality: Validate all critical indicators for accuracy, completeness, consistency, and suitability for analysis.
- Dashboard development: Complete all five planned dashboard sections at a minimum viable level.
- Interactivity: Achieve a minimum 95% pass rate across applicable interaction tests, including filters, navigation, area comparisons, and drill-through features where implemented.
- Quality assurance: Resolve all critical defects before final submission.
- Decision support: Present clear comparisons and evidence based insights that can help decision makers identify areas requiring further investigation or potential intervention.

Project Scope

The project covers the complete development of an interactive urban analytics dashboard, including:

- Collecting suitable public or otherwise permitted datasets.
- Preparing, cleaning, integrating, and documenting data from multiple sources.
- Developing analytical models, indicators, measures, and transparent calculations.
- Designing interactive visualizations for exploring urban conditions and comparing geographical areas.
- Incorporating AI-assisted analysis or generated content when it adds value and can be appropriately validated.
- Conducting data-quality checks, functionality testing, and final validation.
- Preparing the final dashboard, supporting documentation, and project presentation.

The dashboard is intended to support evidence based exploration and decision making. Its findings will inform users about observed patterns and potential priorities but will not, by themselves, establish causation or prove the effectiveness of proposed interventions.

2. Project Plan

Task	Start Wk	End Wk	Dependency	Deliverable
Requirements and scope	1	1	—	Requirements baseline
Source research and feasibility	1	2	Requirements	Source inventory and feasibility review
Data acquisition	2	3	Feasibility review	Documented raw datasets
Cleaning and validation	3	4	Acquired data	Prepared datasets and quality log
Data model and KPI development	4	5	Validated core data	Model and validated measures
Exploratory/spatial analysis	5	6	Initial model and KPIs	Supported findings
Power BI dashboard development	5	8	Validated measures	Five-section prototype
Optional scenario analysis	7	8	Suitable baseline and assumptions	Scenario comparison, if feasible
Testing and refinement	8	9	Integrated prototype	Test results and corrected dashboard
Documentation and presentation	8	10	Outputs available progressively	Final report and presentation
Final review and packaging	10	10	Testing and documentation	Submission package

Project Gantt Chart

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10
Requirements	■									
Feasibility	■	■								
Acquisition		■	■							
Cleaning			■	■						
Model/KPIs				■	■					
Analysis					■	■				
Dashboard					■	■	■	■		
Scenarios (optional)							■	■		
Testing								■	■	
Docs/presentation								■	■	■
Final delivery										■

Milestones:

W1—requirements approved

W2—data feasibility gate

W4—prepared data reviewed

W5—model and core KPIs validated

W8—five section dashboard draft completed

W9—quality assurance completed

W10—final package ready

The critical path is feasibility → acquisition → cleaning → modeling/KPIs → dashboard completion → testing → final delivery. Optional scenarios will be removed if they threaten core deliverables.

Resources:

Power BI Desktop, spreadsheet tools, Power query, SQL, Python, shared repository, and a task tracker are the planned core resources; no paid data service is assumed.

3. Task Assignment & Roles

Proposed role	Defined responsibilities
Project Manager / Team Coordinator	Owns scope, schedule, task tracking, risk review, decisions, and final integration.
Data Research and Acquisition Analyst	Finds candidate datasets; checks source reliability, dates, geography, licensing, and records provenance.
Data Cleaning and Transformation Analyst	Cleans and standardizes data, checks missingness and duplicates, validates fields and coordinates, and documents transformations.
Data Analyst and KPI Developer	Defines indicators, develops calculations, performs exploratory analysis, and validates findings.
Power BI Dashboard Developer	Builds the model implementation, dashboard pages, filters, navigation, and interactions.
QA and Documentation Lead	Maintains test cases and defect logs, checks calculations and interactions, and completes technical documentation.
Presentation and Reporting Lead	Prepares the final presentation, report integration, and dashboard demonstration.

Activity	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
Requirements and scope	Project Manager	Project Manager	All role leads	Team
Source feasibility/acquisition	Data Research Analyst	Project Manager	Data Analyst, GIS if needed	Team
Cleaning and validation	Transformation Analyst	Project Manager	Research and Data Analysts	Team
Model, KPIs, and analysis	Data Analyst	Project Manager	Transformation and Power BI roles	Team
Dashboard development	Power BI Developer	Project Manager	Data Analyst, GIS if needed	Team
Testing and documentation	QA Lead	Project Manager	All technical roles	Team
Presentation and final delivery	Presentation Lead / Project Manager	Project Manager	QA and technical roles	Team

Coordination will use a weekly meeting, a shared task tracker, a decision log, and a common file repository. Raw data will be kept separate from prepared data, file versions will be named consistently, and one designated owner will integrate the final Power BI file. Delays will be handled by identifying affected dependencies, reallocating work where possible, and deferring optional features before reducing testing or documentation.

4. Risk Assessment & Mitigation Plan

Risks are scored as Likelihood × Impact, with each factor rated from 1 (low) to 5 (very high). Scores of 1–4 are Low, 5–9 Medium, 10–16 High, and 17–25 Critical. Scores are initial planning estimates and will be reviewed weekly.

ID	Risk	Score / Priority	Early warning	Prevention	Contingency
R1	Insufficient or inaccessible urban data	5×5=25 Critical	Core indicators lack suitable sources by W2	Run an early feasibility review; document source access and fields	Narrow or omit the indicator; use a defensible alternative
R2	Missing, outdated, inconsistent data	4×5=20 Critical	High missingness or conflicting totals	Profile sources and document dates, definitions, and limitations	Exclude unreliable fields or qualify the analysis
R3	Incompatible geographic boundaries/resolution	4×4=16 High	Areas cannot be compared consistently	Check boundaries, coordinates, and geographic granularity early	Use a common defensible geography or omit the comparison
R4	No population, demand, capacity, or employment data	5×4=20 Critical	No valid denominator or operational measure	Check required fields before committing to ratios or demand claims	Use counts/distribution only with clear limitations
R5	Schedule pressure and academic commitments	5×4=20 Critical	Repeated missed tasks or reduced availability	Track weekly progress and protect testing time	Prioritize core pages and defer optional features
R6	Incorrect calculations or misleading interpretation	3×5=15 High	Measures disagree with independent checks	Use KPI definitions, peer review, and evidence-linked conclusions	Correct or remove affected visuals and rerun tests
R7	Unequal contributions or communication gaps	3×4=12 High	Unowned tasks or unresolved blockers	Assign owners and review the task tracker weekly	Reallocate work and escalate unresolved issues
R8	File corruption, performance, or version conflicts	3×4=12 High	Conflicting PBIX files or slow interactions	Back up versions and test incrementally	Restore last valid version and simplify the model
R9	Scope creep or unsupported scenario analysis	4×4=16 High	New features threaten core milestones	Use change control; require a reliable baseline for scenarios	Defer scenarios and return to minimum viable scope
R10	Licensing, privacy, or incomplete final testing	3×5=15 High	Unclear reuse conditions or open critical defects	Check licenses and attribution; maintain test and delivery checklists	Exclude questionable data and resolve critical defects before submission

5. KPIs (Key Performance Indicators)

The following KPIs measure project delivery and prototype quality. Targets are proposed benchmarks; results will be recorded only after measurement. Production system uptime, actual government response time, and real-world user adoption are excluded because the dashboard is not deployed.

Category	KPI	Calculation / Definition	Proposed target	Frequency, source, owner
Project management	Milestone completion rate	Milestones completed ÷ milestones due × 100	100% of milestones due	Weekly; task tracker; Project Manager
Project management	On-time task completion	Tasks completed by due date ÷ tasks due × 100	≥90%	Weekly; task tracker; Project Manager
Data quality	Source verification rate	Datasets with verified provenance/access/coverage ÷ datasets used × 100	100%	At acquisition and final review; source inventory; Research Analyst
Data quality	Data completeness	Valid non-missing required values ÷ expected required values × 100	≥95% for critical fields where feasible	Each cleaning cycle; data-quality log; Transformation Analyst
Data quality	Indicator documentation coverage	Indicators with definition, unit, date, source, and limitations ÷ indicators implemented × 100	100%	At KPI creation and final review; KPI dictionary; Data Analyst
Analytical quality	Indicator validation rate	Implemented indicators passing validation ÷ indicators tested × 100	100%	At model changes and final review; validation log; Data Analyst
Analytical quality	Evidence-supported conclusions	Conclusions linked to documented evidence ÷ formal conclusions × 100	100%	Analysis review; findings register; Data Analyst
Dashboard quality	Required page completion	Accepted required pages ÷ 5 planned sections × 100	100%	Weekly during development; page checklist; Power BI Developer
Dashboard quality	Interaction test pass rate	Passed interaction tests ÷ tests executed × 100	≥95%; all critical tests pass	QA cycles; test register; QA Lead
Dashboard quality	Open critical defects	Count of unresolved critical defects	0 at submission	Weekly from W8; defect log; QA Lead
Decision-support value	Evidence-based priority areas	Count of areas meeting documented priority criteria	Target ≥3 only if data support comparison	Analysis review; findings register; Data Analyst
Decision-support value	Recommendation evidence linkage	Recommendations with evidence and assumptions ÷ recommendations × 100	100%	Final analysis review; recommendation register; Data Analyst
Final delivery	Documentation completion	Completed required documentation items ÷ total required items × 100	100%	Weekly from W8; checklist; QA Lead
Final delivery	On-time submission	Actual submission compared with confirmed deadline	On or before official deadline	At submission; submission record; Project Manager

Performance will be summarized in a final KPI record with the target, measured result, date, evidence source, and status. Items not yet measured will be marked accordingly; scenario-related indicators will be marked not applicable if the feature is not implemented.